

Question-4

Part-A: Configuration of Knockd

- First, I set my network adaptor to host-only mode to allow the VM and my local pc to be on the same network.
- Next, I installed the knockd on the VM using the command: `sudo apt install knockd`.
- Next, I opened the knockd config file, which had the following default configurations:-

```
GNU nano 7.2 /etc/knockd.conf
[options]
UseSyslog

[openSSH]
sequence    = 7000,8000,9000
seq_timeout = 5
command     = /sbin/iptables -A INPUT -s %IP% -p tcp --dport 22 -j ACCEPT
tcpflags    = syn

[closeSSH]
sequence    = 9000,8000,7000
seq_timeout = 5
command     = /sbin/iptables -D INPUT -s %IP% -p tcp --dport 22 -j ACCEPT
tcpflags    = syn

[openHTTPS]
sequence    = 12345,54321,24680,13579
seq_timeout = 5
command     = /usr/local/sbin/knock_add -i -c INPUT -p tcp -d 443 -f %IP%
tcpflags    = syn
```

- I changed this to below:-

```
[options]
UseSyslog
Interface = enp0s3

[openSSH]
sequence    = 5123,5456,5789
seq_timeout = 20
command     = /usr/sbin/iptables -I INPUT -s %IP% -p tcp --dport 22 -j ACCEPT
tcpflags    = syn

[closeSSH]
sequence    = 6123,6456,6789
seq_timeout = 20
command     = /usr/sbin/iptables -D INPUT -s %IP% -p tcp --dport 22 -j ACCEPT
tcpflags    = syn
```

- Here, I first set up the network interface to be used as “enp0s3” which represents the network adaptor that is in host-only mode.

- Next, I changed the ports required to open the hidden SSH port (22) to this: 5123,5456,5789.
- Here, for demo purposes, I have kept the seq_timeout to 20; otherwise, it should be between 5 and 10 for better security.
- Next, I gave the command that should be invoked when these ports are knocked sequentially, which is: `iptables -I INPUT -s %IP% -p tcp --dport 22 -j ACCEPT`

Through this command, we are basically adding an iptable rule to the INPUT chain, and this rule is added at the top of the chain due to the `-I` flag. Through this rule, we simply allow access to the SSH port for a particular IP address.

- Then, I gave the tcpflags as syn, which represents connection request flags.
- Hence, this means there should be three tcp connection establishment requests sent to these ports sequentially: 5123,5456,5789, in order to open the SSH port for that IP address only.
- The other is the closing of the SSH port, which has ports as 6123,6456,6789, and the main difference is that here, we delete the rule we have applied earlier by passing the flag `-D`.
- Next, I enabled the knockd service by setting the flag `START_KNOCKD = 1` in the file: `/etc/default/knockd` and adding the name of the network interface.

Below is the ss for the same.

```
GNU nano 7.2 /etc/default/knockd
# control if we start knockd at init or not
# 1 = start
# anything else = don't start
# PLEASE EDIT /etc/knockd.conf BEFORE ENABLING
START_KNOCKD=1

# command line options
KNOCKD_OPTS="-i enp0s3"
```

- Now, initially, I added an iptable rule to block all the requests for ssh on port 22 to demonstrate it working.

```

nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 22 -j DROP
nikhil@nikhil-linux:~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination            tcp dpt:ssh
DROP      tcp  --  anywhere              anywhere               tcp dpt:ssh

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
nikhil@nikhil-linux:~$ █

```

- Now, this ensures that initially, all the traffic to port 22 with tcp is blocked. Now, in the knockd config, I have added the iptable command using the -I flag. This ensures that the new access command is on top of the table, and the reason for this is that the iptables rules are matched at the top of the table.
- Then I started the knockd service.

```

nikhil@nikhil-linux:~$ sudo systemctl restart knockd
nikhil@nikhil-linux:~$ sudo systemctl status knockd
● knockd.service - Port-Knock Daemon
   Loaded: loaded (/usr/lib/systemd/system/knockd.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-02-22 00:59:57 IST; 9s ago
     Docs: man:knockd(1)
    Main PID: 4023 (knockd)
      Tasks: 1 (limit: 4611)
     Memory: 552.0K (peak: 880.0K)
        CPU: 20ms
    CGroup: /system.slice/knockd.service
            └─4023 /usr/sbin/knockd -i enp0s3

Feb 22 00:59:57 nikhil-linux systemd[1]: Started knockd.service - Port-Knock Daemon.
Feb 22 00:59:57 nikhil-linux knockd[4023]: starting up, listening on enp0s3
nikhil@nikhil-linux:~$ █

```

- Now I send a series of TCP packets using nmap and then checked the iptable and status of knockd. Below is the ss, which confirms that this worked.

```

C:\Users\DELL>nmap -Pn -p 5123 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-02-22 01:10 India Standard Time
Nmap scan report for 192.168.56.101
Host is up (0.00088s latency).

PORT      STATE SERVICE
5123/tcp  closed unknown
MAC Address: 08:00:27:03:C0:96 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.77 seconds

C:\Users\DELL>nmap -Pn -p 5456 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-02-22 01:10 India Standard Time
Nmap scan report for 192.168.56.101
Host is up (0.0010s latency).

PORT      STATE SERVICE
5456/tcp  closed apc-5456
MAC Address: 08:00:27:03:C0:96 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.74 seconds

C:\Users\DELL>nmap -Pn -p 5789 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-02-22 01:10 India Standard Time
Nmap scan report for 192.168.56.101
Host is up (0.00013s latency).

PORT      STATE SERVICE
5789/tcp  closed unknown
MAC Address: 08:00:27:03:C0:96 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.75 seconds

```

- Knockd status and iptables

```

Feb 22 00:59:57 nikhil-linux systemd[1]: Started knockd.service - Port-Knock Daemon.
Feb 22 00:59:57 nikhil-linux knockd[4023]: starting up, listening on enp0s3
Feb 22 01:08:15 nikhil-linux knockd[4023]: 192.168.56.1: closeSSH: Stage 1
Feb 22 01:09:59 nikhil-linux knockd[4023]: 192.168.56.1: openSSH: Stage 1
Feb 22 01:10:08 nikhil-linux knockd[4023]: 192.168.56.1: openSSH: Stage 2
Feb 22 01:10:15 nikhil-linux knockd[4023]: 192.168.56.1: openSSH: Stage 3
Feb 22 01:10:15 nikhil-linux knockd[4023]: 192.168.56.1: openSSH: OPEN SESAME
Feb 22 01:10:15 nikhil-linux knockd[4087]: openSSH: running command: /usr/sbin/iptables -I INPUT -s 192.168.56.1 -p tcp --dport
nikhil@nikhil-linux:~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target    prot opt source                destination            tcp dpt:ssh
ACCEPT    tcp  --  192.168.56.1          anywhere               tcp dpt:ssh
DROP      tcp  --  anywhere              anywhere               tcp dpt:ssh

Chain FORWARD (policy ACCEPT)
target    prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target    prot opt source                destination
nikhil@nikhil-linux:~$

```

- Next, I was able to access the VM via SSH.

```
C:\Users\DELL>ssh nikhil@192.168.56.101
nikhil@192.168.56.101's password: 
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.11.0-17-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

6 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Wed Feb 19 20:06:19 2025 from 192.168.56.1
nikhil@nikhil-linux:~$ pwd
/home/nikhil
nikhil@nikhil-linux:~$
```

Part-B: Why TCP should be preferred over UDP here

- TCP is a connection-oriented protocol, and it ensures the reliable delivery of the knock packets to the server.
- Using TCP ensures that packets are delivered in order, which is important here, as we are sequentially accessing the ports in this case.

Part-C: Default Choice of Ports for knockd configuration

- The default choice of ports in the knockd configuration is 7000,8000,9000 for opening port 22, and for closing, it has the same ports but in reversed order.

Below is the ss for the default options:-

```
GNU nano 7.2 /etc/knockd.conf
[options]
    UseSyslog

[openSSH]
    sequence      = 7000,8000,9000
    seq_timeout   = 5
    command       = /sbin/iptables -A INPUT -s %IP% -p tcp --dport 22 -j ACCEPT
    tcpflags      = syn

[closeSSH]
    sequence      = 9000,8000,7000
    seq_timeout   = 5
    command       = /sbin/iptables -D INPUT -s %IP% -p tcp --dport 22 -j ACCEPT
    tcpflags      = syn

[openHTTPS]
    sequence      = 12345,54321,24680,13579
    seq_timeout   = 5
    command       = /usr/local/sbin/knock_add -i -c INPUT -p tcp -d 443 -f %IP%
    tcpflags      = syn
```

- These ports are not safe in the regard that these are publicly available as the default options; hence, the attacker also knows these default ports, and these are very common to guess if an attacker discovers this sequence, which he may guess or look into the default options for this, then they can open the SSH service.

Using these ports is analogous to the default password for the router provided by the ISP, as these are easy to guess, and an attacker can exploit these very easily.

To secure this, we can use not-so-common ports and have a longer sequence of ports to be knocked.

References

- Linux Man Page for Knockd: <https://linux.die.net/man/1/knockd>